

ELECTRIC CIRCUITS

PART 3 COMPONENT BEHAVIOUR

LESSONS 8–10

I–V graphs, the I–V practical, and sensors: thermistors and LDRs. Learn with it in class and revise from it at home.

- 8. I–V graphs
- 9. Investigating I–V characteristics
- 10. Thermistors, LDRs and sensors

ANSWER EDITION · FOR TEACHERS AND SELF-MARKING



NAME

CLASS

TEACHER

TARGET GRADE

HOW THIS BOOKLET WORKS

This is **Part 3** of the Electric Circuits workbook. It builds on the parts before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

 DO NOW Quick questions from earlier lessons. Remembering again is what makes it stick.	 LEARN IT The explanation, diagrams and worked examples. Read this again when you revise.	 TRY IT Practice that builds up: first with support, then on your own.
 LAB Record measurements from a virtual lab or a required practical.	 EXAM CORNER An AQA-style question with a tip on how the marks are given.	 REVISE IT Summary, cover-and-answer questions and an “I can” check to colour in.

! **Revising at home?** For each lesson: (1) read the Learn it pages, (2) cover them and answer the “Cover the page” questions in Revise it, (3) check with the quick answers at the back, (4) do the matching mission on the companion website, which has more questions with answers.

THE ROUTE THROUGH THE UNIT · THIS PART IN BOLD

#	LESSON	COMPANION WEBSITE
01	Complete circuits and symbols What a circuit needs, circuit symbols and drawing rules, measuring current	Mission 1 · Build & measure
02	Current and charge Current as a rate of flow of charge, $Q = I t$, current is not used up	Mission 2 · Charge on the move
03	Potential difference p.d. as energy per coulomb, voltmeters, $E = Q V$, p.d.s add in series	Mission 3 · Energy per coulomb
04	Resistance and $V = I R$ Resistance, $V = I R$, ohmic conductors, why heating raises resistance	Mission 4 · Resistance & $V = I R$
05	Resistance of a wire REQUIRED PRACTICAL Required practical: how the length of a wire affects its resistance	Required practical · Resistance
06	Series and parallel circuits Current, p.d. and resistance rules for series and parallel circuits	Mission 5 · Series & parallel
07	Resistors in series and parallel REQUIRED PRACTICAL Required practical: measure total resistance of two resistors in series and in parallel	Required practical · Resistance (Part B)
08	I–V graphs I–V graphs of a fixed resistor, filament lamp and diode	Mission 6 · I–V characteristics
09	Investigating I–V characteristics REQUIRED PRACTICAL Required practical: measure the I–V characteristics of a resistor, lamp and diode	Required practical · I–V characteristics
10	Thermistors, LDRs and sensors How thermistors and LDRs respond to temperature and light, and sensor circuits	Mission 6 · Thermistor and LDR
11	Power and energy transfer Power, $E = P t$, $P = V I$, $P = I^2 R$, energy transfers in appliances	Mission 7 · Power & the National Grid
12	Mains, safety and the National Grid ac and dc, 230 V 50 Hz mains, three-core cable, earthing, the National Grid	Missions 7 and 8

Companion website: panphy.app/gcsephy/year10phy/unit01 · missions, interactive equation triangles, required-practical pages and the Exam Zone.

EQUATION TOOLKIT

The equations you have met so far. You will meet these one at a time. Learn each equation **in words** as well as symbols: your exams may or may not give you an equation sheet, and knowing them by heart makes every question faster.

$$Q = I t$$

charge flow = current × time

C, A, s

Rearranged: $I = Q \div t$ · $t = Q \div I$

Lesson 2

$$E = Q V$$

energy transferred = charge × p.d.

J, C, V

Rearranged: $V = E \div Q$ · $Q = E \div V$

Lesson 3

$$V = I R$$

p.d. = current × resistance

V, A, Ω

Rearranged: $I = V \div R$ · $R = V \div I$

Lesson 4

$$R_{\text{total}} = R_1 + R_2$$

total resistance in series

Ω

Rearranged: (series only)

Lesson 6

FOUR STEPS FOR EVERY CALCULATION

1

Write down what you know, with units. Convert units first (minutes → seconds, kW → W, mA → A).

2

Write the equation you will use, in symbols.

3

Substitute the numbers, then rearrange if needed.

4

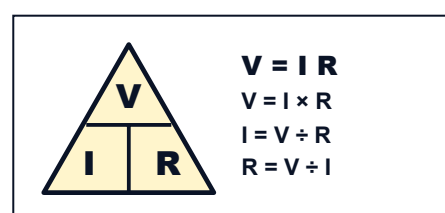
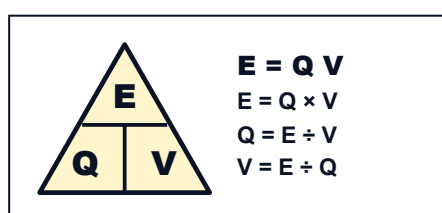
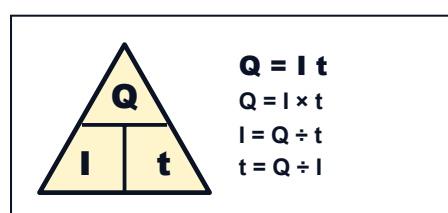
Give the answer with a unit. Ask: is it sensible?

CONVERT	HOW	EXAMPLE
minutes → seconds	× 60	3.0 min = 180 s
hours → seconds	× 3600	2.0 h = 7200 s
milliamps (mA) → amps (A)	÷ 1000	250 mA = 0.25 A
kilowatts (kW) → watts (W) · kilohms (k Ω) → ohms (Ω)	× 1000	2.2 kW = 2200 W
kilovolts (kV) → volts (V)	× 1000	400 kV = 400 000 V

EQUATION TRIANGLES: HELP WITH REARRANGING

- 1 Cover the quantity you want to find.
- 2 The two left **side by side**? **Multiply** them.
- 3 One **above** the other? **Divide** the top by the bottom.

Example: to find I from $Q = I t$, cover I. Q is above t, so $I = Q \div t$. Triangles only help with rearranging: you still need to learn each equation.



Interactive version: panphy.app/gcsephy/year10phy/unit01/equation-triangles.

BIG QUESTION Every component has a 'fingerprint': the shape of its I-V graph. Could you identify a component from its graph alone?

BY THE END YOU CAN

- read an I-V graph, including negative values
- sketch and recognise the graphs for a fixed resistor, a filament lamp and a diode
- calculate the resistance at a point with $R = V \div I$
- explain the shape of each graph

I-V characteristic

filament lamp

diode

LED

forward

reverse

ONLINE COMPANION

Mission 6 · I-V characteristics

Revision notes with graphs and 5 practice questions.

panphy.app/gcsephy/year10phy/unit01/lesson/iv-characteristics/



RETRIEVE

DO NOW

From earlier lessons. No notes!

1

What is an ohmic conductor?

Current $\propto$ p.d. at constant

temperature (constant

resistance).

2

Why does a metal's resistance increase when it gets hot?

Ions vibrate more, so more

collisions.

3

Calculate R for 4.0 V and 0.20 A.

20 Ω



EXPLAIN

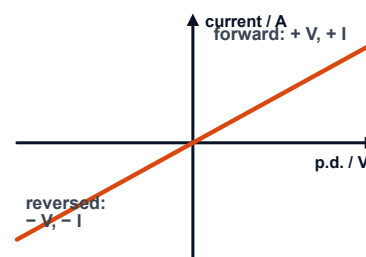
READING COMPONENT FINGERPRINTS

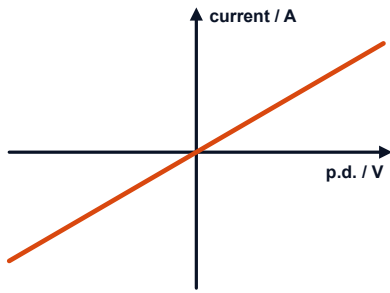
On an **I-V graph**, current (I) goes up the side and p.d. (V) goes along the bottom.

Negative values mean the component is connected **the other way round**, so the current flows the other way.

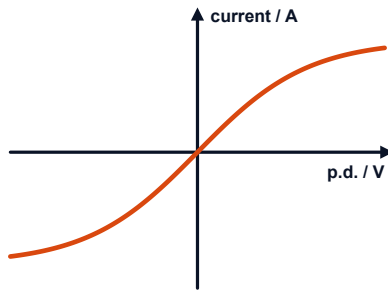


To find the resistance at any point, read V and I and use $R = V \div I$. Do not use the gradient.

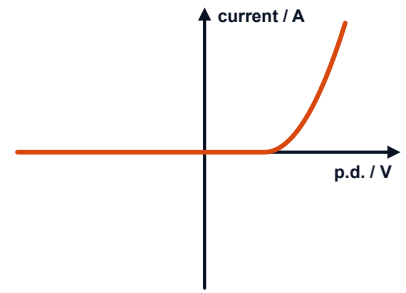




Fixed resistor: straight line through the origin. Constant resistance (at constant temperature).



Filament lamp: gets less steep. Resistance increases as the filament heats up.



Diode: current flows in one direction only. Very high resistance in reverse.

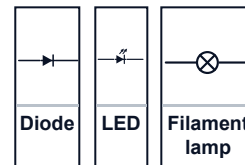
WHY DOES THE LAMP GRAPH BEND?

As the p.d. increases, the current increases. The current **heats the filament**. Hotter metal ions vibrate more, so electrons collide with them more often and the **resistance increases**. Each extra volt adds less current than the one before, so the graph gets less steep. It has the same shape for negative p.d. because the filament heats up whichever way the current flows.

DIODES AND LEDS

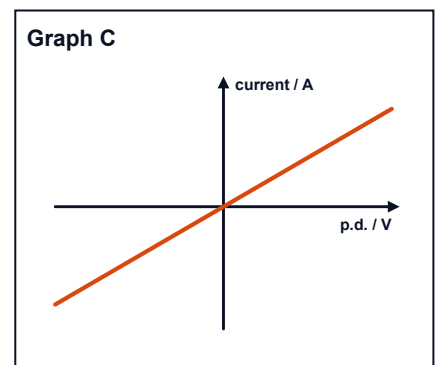
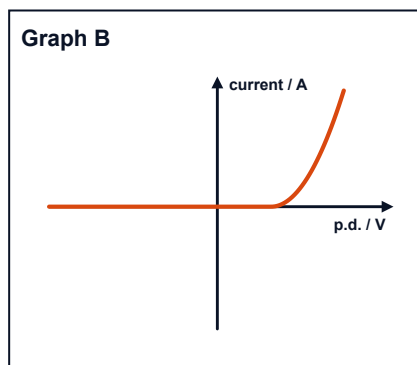
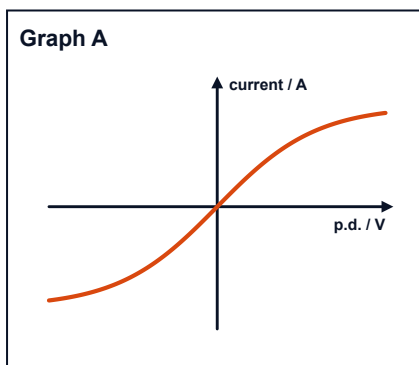
A **diode** lets current flow in one direction only: the way its triangle points. In the other direction its resistance is very high, so there is almost no current. Diodes protect circuits, for example if a battery is put in the wrong way round.

An **LED** (light-emitting diode) is a diode that gives out light when a current flows through it.



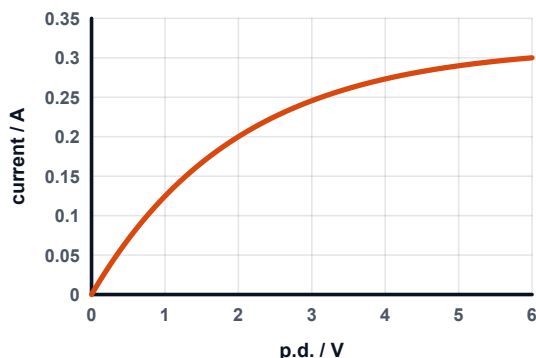
PRACTISE TRY IT

1 Name the component that gives each graph.



A: filament lamp. B: diode. C: fixed resistor.

- 2 This is the I–V graph for a filament lamp. (a) Read the current at 2.0 V and at 6.0 V. (b) Calculate the resistance at each p.d. (c) Explain why they are different.



(a) 0.20 A and 0.30 A. (b) $R = 2.0 \div 0.20 = 10 \, \Omega$; $R = 6.0 \div 0.30 = 20 \, \Omega$. (c) The larger current heats the filament more, so its resistance increases.

- 3 Number these statements 1–6 to explain the shape of a filament lamp's graph.

4 | The electrons collide with the ions more often

2 | The filament gets hotter

6 | So the I–V graph gets less steep

1 | The p.d. and the current increase

3 | The metal ions vibrate more

5 | The resistance of the filament increases

- 4 A component carries 0.10 A at 2.0 V and 0.15 A at 4.0 V. Is it an ohmic conductor? Show your working.

$R = 2.0 \div 0.10 = 20 \, \Omega$; $R = 4.0 \div 0.15 = 27 \, \Omega$. The resistance changes, so it is not ohmic.

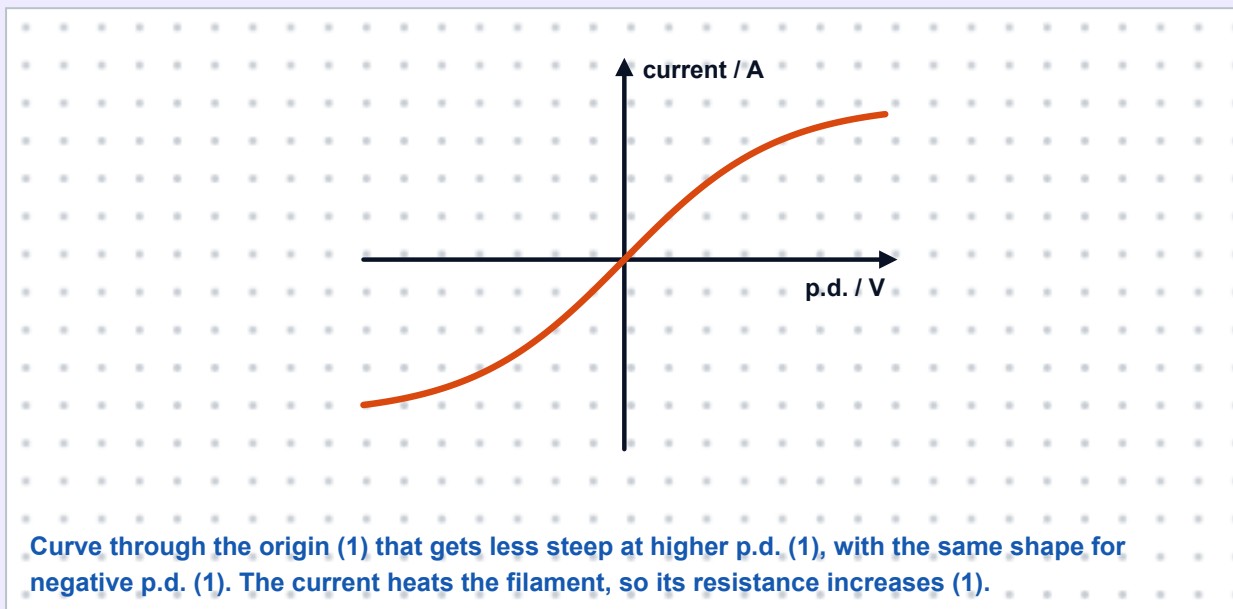
- 5 What happens to the current through a diode when it is connected the other way round? Suggest one use for this.

There is (almost) no current: the diode has a very high resistance in reverse. Use: protecting a circuit from a battery put in backwards, or making sure current flows only one way.



Get the marks: A sketch needs labelled axes and the right shape in both directions. The explanation needs cause → effect: current → heating → resistance.

- 6 Sketch the I–V graph of a filament lamp for positive and negative p.d. Then explain its shape. [4 marks]

**SUMMARY**

- I–V graph: current up the side, p.d. along the bottom. Negative = connected the other way.
- Fixed resistor: straight line through the origin (constant R).
- Filament lamp: gets less steep; the filament heats up, so R increases.
- Diode: current in one direction only; very high resistance in reverse. An LED is a diode that gives out light.
- Resistance at a point: $R = V \div I$.

COVER THE PAGE AND ANSWER

1 Sketch the I–V graph for a fixed resistor.

2 Why does a filament lamp's resistance increase?

3 What does a diode do?

4 How do you find R at a point on an I–V graph?

I CAN...

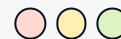
read values from an I–V graph, including negative values



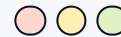
sketch the I–V graphs for a fixed resistor, lamp and diode



calculate resistance at a point



explain why the lamp graph curves



INVESTIGATING I-V CHARACTERISTICS

AQA Combined RP 16 · Physics RP 4

BIG QUESTION Can you produce the fingerprint graphs of a resistor, a lamp and a diode yourself?

BY THE END YOU CAN

- set up a circuit to vary the p.d. and measure the current
- take readings in both directions, safely
- plot an I-V graph with positive and negative values
- describe how the method changes for a diode

variable resistor

protective resistor

reverse

range

interval

ONLINE COMPANION

Required practical · I-V characteristics

Method, example results, graphs and a printable worksheet.

panphy.app/gcsephy/year10phy/unit01/practical/iv-characteristics/



RETRIEVE

DO NOW

From earlier lessons. No notes!

1

Sketch the I-V graph shape for a diode.

Zero current for negative p.d.;
almost none until a small
positive p.d., then rises steeply.

2

Why does a lamp's resistance increase as the current increases?

The filament heats up, so more
collisions.

3

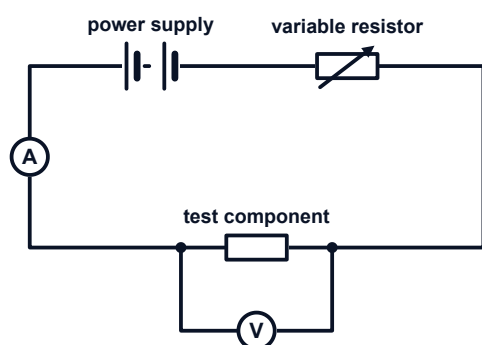
How is a voltmeter connected to a component?

Across it, in parallel.

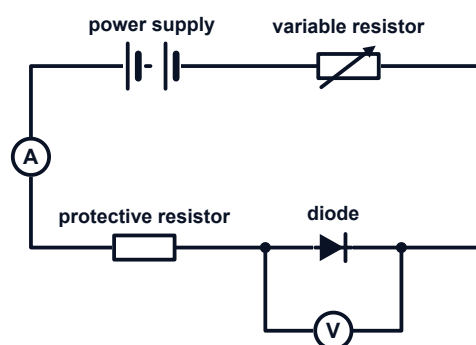


EXPLAIN

PLAN THE PRACTICAL



Resistor first, then swap in the filament lamp.



Diode: add a protective resistor; voltmeter across the diode only.

The **variable resistor** changes the p.d. across the test component. To get negative values, **swap the connections** to the power supply so the current flows the other way, and record those readings as negative.

- 1 Set up the circuit with the fixed resistor as the test component.
- 2 Use the variable resistor to set a range of p.d.s, for example from 0 to 6 V. Record V and I each time.

- 3 Reverse the connections to the power supply. Repeat, recording V and I as negative values.
- 4 Repeat for the filament lamp.
- 5 Repeat for the diode with its protective resistor, taking small steps of p.d. from 0 to about 1 V.

STAY SAFE

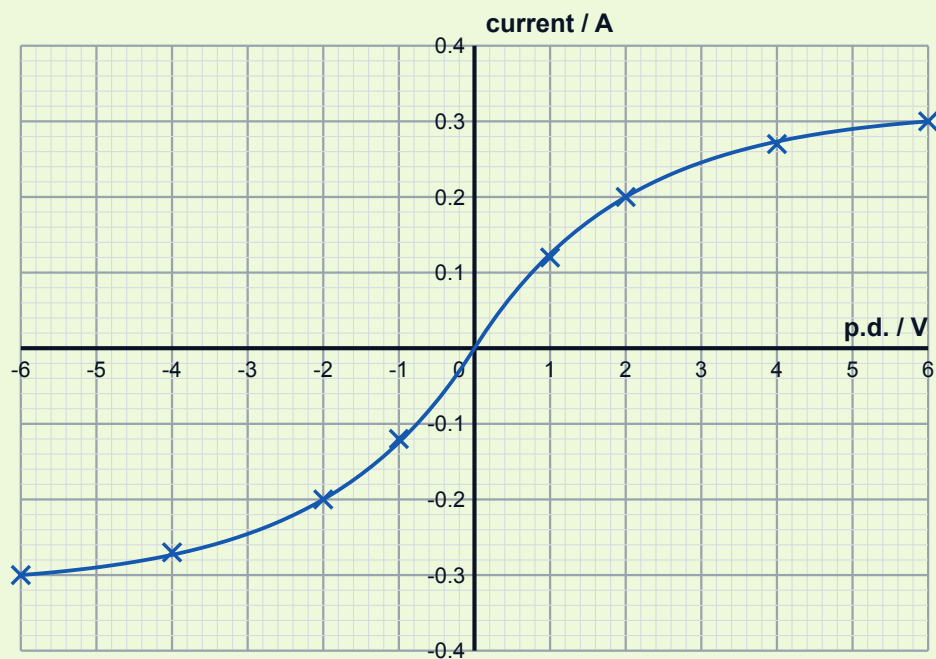
- The lamp gets hot. Do not touch it while it is on or just after.
- Do not go above the lamp's rated p.d. (for example 6 V).
- Always use the protective resistor with the diode, so the current stays small.



- 6 Aim for these p.d.s. Record the actual p.d. if it is slightly different.

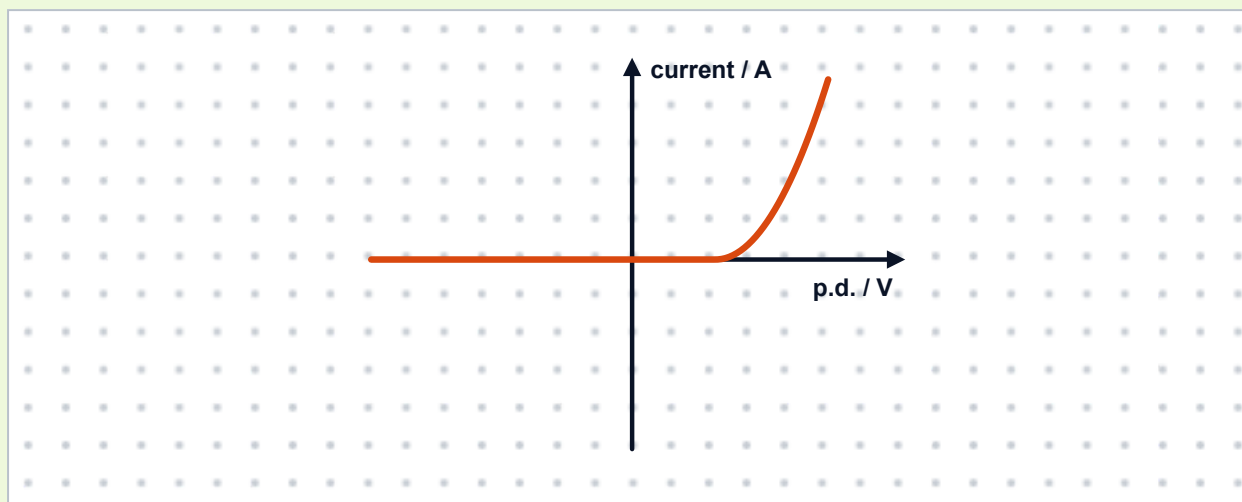
p.d. / V	Resistor: current / A	Lamp: current / A
-6.0	-0.30	-0.30
-4.0	-0.20	-0.27
-2.0	-0.10	-0.20
-1.0	-0.05	-0.12
0	0	0
1.0	0.05	0.12
2.0	0.10	0.20
4.0	0.20	0.27
6.0	0.30	0.30

- 7 Plot both components on the same axes. Use a different symbol for each, and add a key.



8 Diode (with protective resistor).

p.d. / V	-1.0	0	0.2	0.4	0.5	0.6	0.65	0.7
current / mA	0	0	0	0	0.5	3.0	8.0	20



Sketch the shape of the diode graph from your readings.



PRACTISE

ANALYSE AND EVALUATE

- 1 Describe the shape of your fixed-resistor graph. What does it show?

A straight line through the origin, the same in both directions. Current $\propto$ p.d., so the resistance is constant.

- 2 Use your lamp results to calculate its resistance at 1.0 V and at 6.0 V. Explain the difference.

(Example) $1.0 \div 0.12 = 8.3 \Omega$; $6.0 \div 0.30 = 20 \Omega$. At a higher current the filament is hotter, so its resistance is higher.

- 3 Why is a protective resistor used with the diode? Tick **one** box.

☐

to make the diode light up

☒

to limit the current so the diode is not damaged

☐

to measure the p.d.

☐

to reverse the current

- 4 Why were the diode readings taken in smaller steps of p.d.?

The current changes very quickly over a small range of p.d. (about 0.5–0.7 V), so small steps are needed to show the shape.

- 5 A student plots the reversed readings as positive numbers. What will be wrong with the graph?

All the points would be in the top-right quadrant. Reversed readings should be negative V and negative I, in the bottom-left quadrant.



Get the marks: Name the equipment and say what it is for. Cover both directions, safety, and the changes for a diode.

- 9** Describe a method to obtain the I–V characteristic of a filament lamp for positive and negative p.d. State two ways the method would change for a diode. **[6 marks]**

Any six: ammeter in series and voltmeter across the lamp; vary the p.d. with a variable resistor or variable supply; record pairs of V and I readings over a range; reverse the supply connections to get negative values; stay below the lamp's rated p.d.; plot I against V. Diode: add a protective resistor in series; use smaller p.d. steps / a milliammeter.

**SUMMARY**

- Vary the p.d. with a **variable resistor**; ammeter in series, voltmeter across the component.
- Reverse the supply connections for negative readings, and record them as negative.
- Diode: protective resistor, a milliammeter and small steps of p.d.
- Plot I (up) against V (along), with the origin in the middle.

COVER THE PAGE AND ANSWER

1 How do you change the p.d. across the component?

2 How do you get negative readings?

3 Why is a protective resistor used with a diode?

4 Where do reversed readings go on the graph?

I CAN...

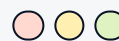
set up the I–V circuit and explain each part



take and record readings in both directions



plot an I–V graph with the origin in the middle



describe the extra steps for a diode



THERMISTORS, LDRS AND SENSORS

AQA Combined 6.2.1.4, 6.2.2 · Physics 4.2.1.4, 4.2.2

BIG QUESTION How does a streetlight know it's getting dark, and how does a fire alarm know the room is hot?

BY THE END YOU CAN

- describe how a thermistor's resistance changes with temperature
- describe how an LDR's resistance changes with light intensity
- predict the current in a sensor circuit
- choose the right sensor for a job

thermistor

LDR

light intensity

sensor

ONLINE COMPANION

Mission 6 · Thermistor and LDR

Read the 'Thermistor' and 'Light-dependent resistor' sections and their questions.

panphy.app/gcsephy/year10phy/unit01/lesson/iv-characteristics/



RETRIEVE

DO NOW

From earlier lessons. No notes!

1

Which component's I–V graph gets less steep?

Filament lamp.

2

At a fixed p.d., what happens to the current if the resistance doubles?

It halves.

3

Draw the symbol for a diode.

A triangle pointing at a bar, on a line.

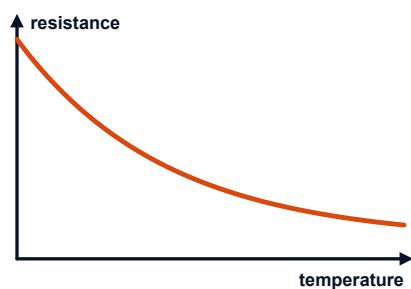
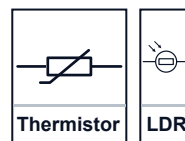


EXPLAIN

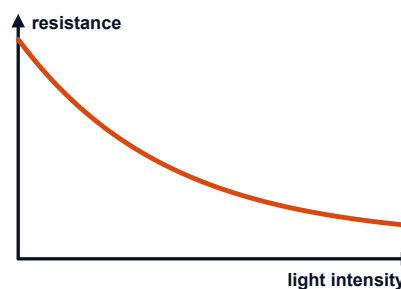
COMPONENTS THAT SENSE

Some resistors change their resistance when their surroundings change. That makes them useful as **sensors**.

A **thermistor** responds to **temperature**. A **light-dependent resistor (LDR)** responds to **light intensity** (how bright it is).



Thermistor: as the temperature increases, its resistance decreases.



LDR: as the light gets brighter, its resistance decreases.



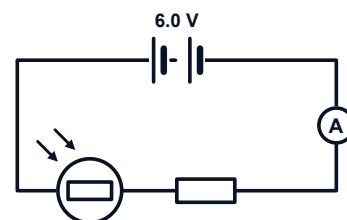
Both follow **more** → **less**: more heat or more light means less resistance.

	THERMISTOR	LDR
Responds to	temperature	light intensity
When that increases...	resistance decreases	resistance decreases
Used in	thermostats, fire alarms, car engine sensors	night lights, streetlights, security lights

SENSORS IN CIRCUITS

In a series circuit with a fixed supply p.d., the current depends on the **total resistance**. When light shines on the LDR, its resistance falls, so the total resistance falls and the **current rises** ($I = V \div R$).

A change in current (or in the p.d. across the fixed resistor) can be used to switch something on, such as a lamp or an alarm.



An LDR in series with a fixed resistor.

WORKED EXAMPLE

A 6.0 V supply is connected to an LDR and a resistor in series. In bright light the total resistance is 600 Ω . Calculate the current.

EQUATION $I = V \div R$

SUBSTITUTE $I = 6.0 \div 600$

ANSWER $I = 0.010 \text{ A (10 mA)}$

YOUR TURN

In the dark, the total resistance rises to 1500 Ω . Calculate the new current.

EQUATION $I = V \div R$

SUBSTITUTE $I = 6.0 \div 1500$

ANSWER $I = 0.0040 \text{ A (4.0 mA)}$



PRACTISE TRY IT

1 Complete the sentences.

When the temperature of a thermistor increases, its resistance decreases. When the light on an LDR gets brighter, its resistance decreases. So in the dark, an LDR has a high resistance.

WORD BANK increases decreases high low

2 Choose a thermistor or an LDR for each job, and give a reason.

Job	Sensor	Why
Fire alarm	thermistor	detects a rise in temperature
Automatic garden light	LDR	detects when it gets dark
Greenhouse heater that switches on when it is cold	thermistor	detects a fall in temperature
Camera that measures how bright a scene is	LDR	detects light intensity

- 3 A thermistor is in series with an ammeter and a 12 V supply. The thermistor is put into hot water. Describe and explain what happens to the ammeter reading.

The reading increases. The thermistor gets hotter, so its resistance decreases, the total resistance decreases, and $I = V \div R$ increases.

- 4 **SUPPORTED** A 12 V sensor circuit has a total resistance of 300 Ω . Calculate the current.

EQUATION	$I = V \div R$	
SUBSTITUTE	$I = 12 \div 300$	
ANSWER	0.040 A	unit:

- 5 Which change would **increase** the current through a thermistor in a series circuit?

- ☐ cooling the thermistor
 ☒ heating the thermistor
 ☐ covering it with a black cloth
 ☐ adding another resistor in series



AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: For 'explain', link each step: light $\rightarrow$ LDR resistance $\rightarrow$ total resistance $\rightarrow$ current.

- 6 A 6.0 V supply is connected to an LDR and a fixed resistor in series. In dim light the total resistance is 1500 Ω . (a) Calculate the current. (b) Explain what happens to the current when the light becomes brighter. **[4 marks]**

(a) $I = V \div R$ (1) = $6.0 \div 1500 = 0.0040$ A (1). (b) The LDR's resistance decreases (1), so the total resistance decreases and the current increases (1).



SUMMARY

- Thermistor: temperature up → resistance **down**.
- LDR: light intensity up → resistance **down**.
- In a series circuit at a fixed p.d.: lower total resistance → bigger current.
- Thermistors sense temperature (thermostats, fire alarms); LDRs sense light (streetlights, night lights).

COVER THE PAGE AND ANSWER

1 What happens to a thermistor's resistance as it warms up?

2 What happens to an LDR's resistance in the dark?

3 Which sensor would you use in a streetlight?

4 Total R falls in a series circuit. What happens to the current?

I CAN...

describe how a thermistor responds to temperature



describe how an LDR responds to light



explain how the current in a sensor circuit changes



choose and justify a sensor for a job



KEY WORDS

Learn the meaning and spelling. In exams, precise words earn marks: say “p.d. across” and “current through”.

Ammeter · Measures current. Connected in series.

Ampere (A) · Unit of current. $1\text{ A} = 1\text{ C per second}$.

Anomaly · A result that does not fit the pattern of the others.

Battery · Two or more cells joined together.

Charge (Q) · Carried by particles such as electrons. Unit: coulomb (C).

Conventional current · The agreed direction of current: from + to –.

Current (I) · The rate of flow of charge.

Diode · Lets current flow in one direction only.

Directly proportional · If one quantity doubles, the other doubles. The graph is a straight line through the origin.

Electron · A tiny negatively charged particle. Free electrons carry the current in metals.

Filament lamp · A lamp with a thin metal filament; its resistance increases as it heats up.

I–V characteristic · A graph of current against p.d. for a component.

LDR · Light-dependent resistor: resistance decreases as light intensity increases.

LED · Light-emitting diode: a diode that gives out light.

Ohm (Ω) · Unit of resistance.

Ohmic conductor · Current is directly proportional to p.d. at constant temperature.

Parallel · Components on separate branches. A voltmeter is connected in parallel.

Potential difference (V) · Energy transferred per unit charge between two points. Unit: volt (V).

Resistance (R) · How much a component opposes the current. Unit: ohm (Ω).

Series · Components connected one after another in a single loop.

Thermistor · Resistance decreases as temperature increases.

Volt (V) · Unit of p.d. $1\text{ V} = 1\text{ J per C}$.

Voltmeter · Measures p.d. Connected in parallel, across a component.

PROGRESS TRACKER

Colour one circle for each statement: **red** = not yet, **amber** = getting there, **green** = I can do this without help. Revisit your reds first.

I CAN...	R · A · G
08 · I–V graphs	
read values from an I–V graph, including negative values	
sketch the I–V graphs for a fixed resistor, lamp and diode	
calculate resistance at a point	
explain why the lamp graph curves	
09 · Investigating I–V characteristics	
set up the I–V circuit and explain each part	
take and record readings in both directions	
plot an I–V graph with the origin in the middle	
describe the extra steps for a diode	
10 · Thermistors, LDRs and sensors	
describe how a thermistor responds to temperature	
describe how an LDR responds to light	
explain how the current in a sensor circuit changes	
choose and justify a sensor for a job	

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising.
Answers to the longer questions are in your teacher's answer edition.

LESSON 8 · I–V GRAPHS

Do now: (1) Current $\propto$ p.d. at constant temperature (constant resistance). (2) Ions vibrate more, so more collisions. (3) $20\ \Omega$
Cover the page: (1) Straight line through the origin. (2) It gets hotter, so the ions vibrate more and there are more collisions. (3) Lets current flow in one direction only. (4) Read V and I; $R = V \div I$.

LESSON 9 · INVESTIGATING I–V CHARACTERISTICS

Do now: (1) Zero current for negative p.d.; almost none until a small positive p.d., then rises steeply. (2) The filament heats up, so more collisions. (3) Across it, in parallel.
Cover the page: (1) Adjust the variable resistor (or variable supply). (2) Swap the connections to the supply. (3) To limit the current and stop damage. (4) In the bottom-left quadrant (negative V and I).

LESSON 10 · THERMISTORS, LDRS AND SENSORS

Do now: (1) Filament lamp. (2) It halves. (3) A triangle pointing at a bar, on a line.
Cover the page: (1) It decreases. (2) It increases (it is high). (3) An LDR. (4) It increases.